NUMERICAL INTEGRATION

Integration of *f(x)* from a to b is $\int_a^b f(x)dx$

- If we divide [a, b] into n subintervals of equal length
- $\Delta x = (b - a)/n$, we have:

$$\int_a^b f(x)dx \approx \sum_{i=1}^{n} f(x_i *)\Delta x$$

- where x_i* is any point in the i_{th} subinterval [$x_{i\ -1}$, x_i].

L_n APPROXIMATION

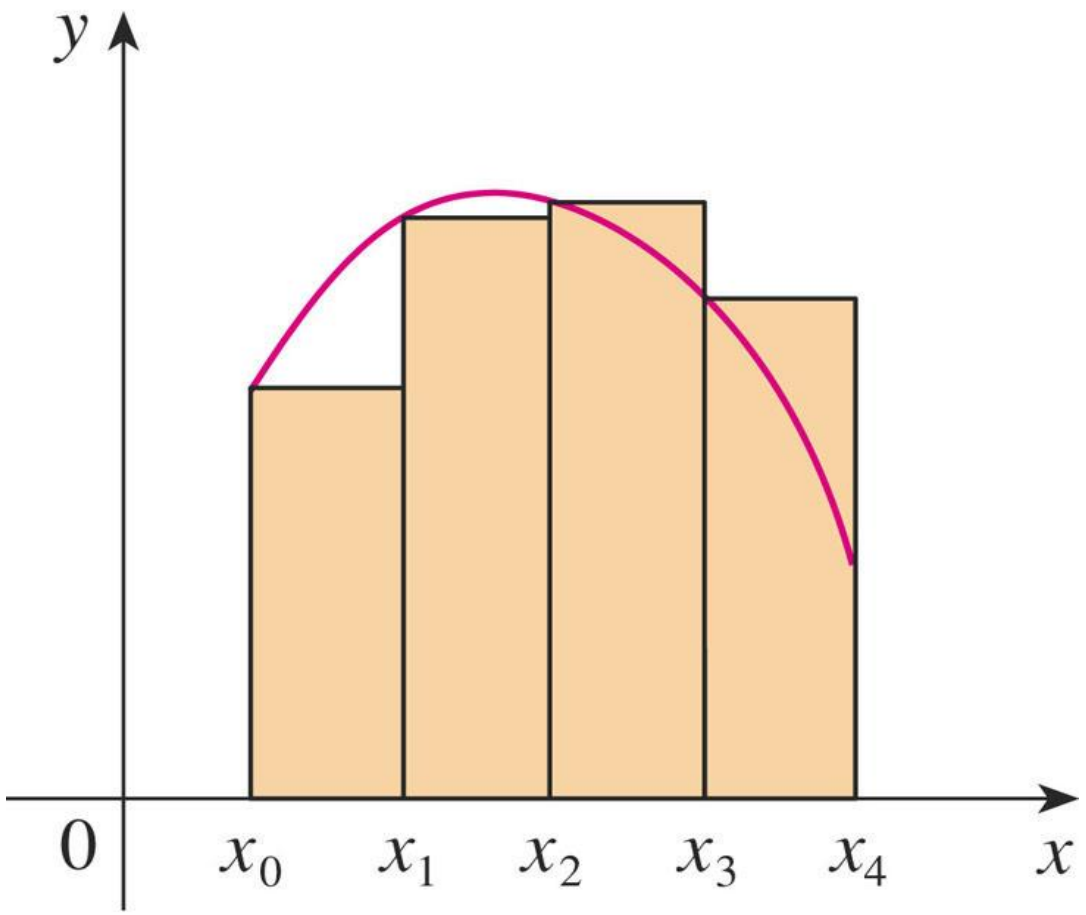
- If x_i* is chosen to be the left endpoint of the interval, then x_i* = $x_{i\ -1}$ and we have:

$$\int_a^b f(x)dx \approx L_n = \sum_{i=1}^{n} f(x_{i-1})\Delta x \qquad \text{Equation-1}$$

- The approximation L_n is called the left endpoint approximation.

L_n APPROXIMATION

- (if $f(x) \geq 0$), $f(x_{i-1}) \times \Delta x$ represents an area of a rectangle, and equation 1, represents the sum of the rectangles, which is an approximation of the area under the curve (shown in Fig (a)).



(a) Left endpoint approximation

APPROXIMATE INTEGRATION

Example 1(a)

- Approximate the integral of $f(x) = 1/x$ with $n = 5$, using left end-point approximation: $\int_1^2 (1/x)dx$

- With $n = 5$, $a = 1$ and $b = 2$, we have: $\Delta x = (2 - 1)/5 = 0.2$
  - <u>So, L_n the left endpoint approximation is :</u>

$$\int_1^2 \frac{1}{x} dx \approx L_5 = [f(1) + f(1.2) + f(1.4) + f(1.6) + f(1.8)] \times 0.2$$

$$= 0.2\left(\frac{1}{1} + \frac{1}{1.2} + \frac{1}{1.4} + \frac{1}{1.6} + \frac{1}{1.8}\right) = 0.745634921 \ (0.693147)$$

APPROXIMATE INTEGRATION

- In Example 1(a), we deliberately chose an integral whose value can be computed explicitly so that we can see how accurate the left end-point approximation is.
  - By the Fundamental Theorem Calculus,

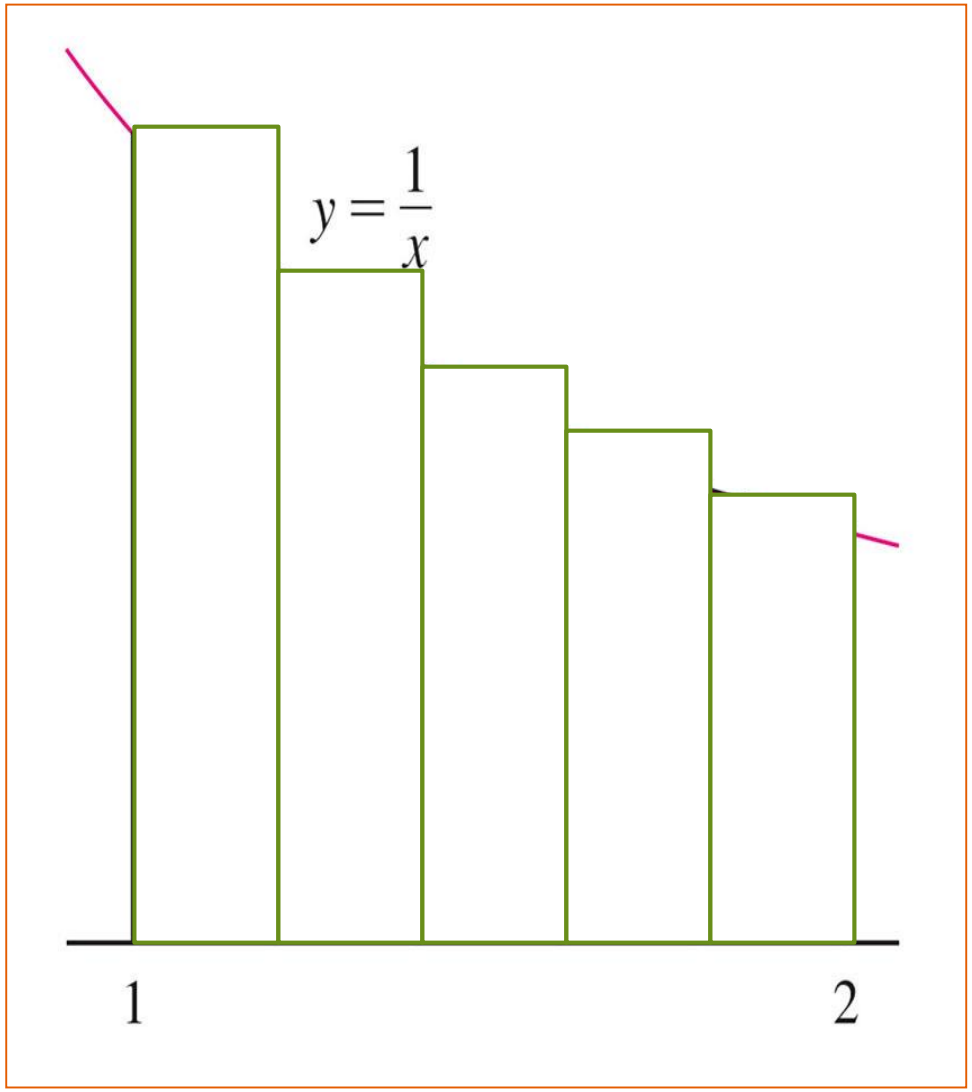
$$\int_1^2 \frac{1}{x}\,dx = \ln x\Big]_1^2 = \ln 2 = 0.693147\ldots$$

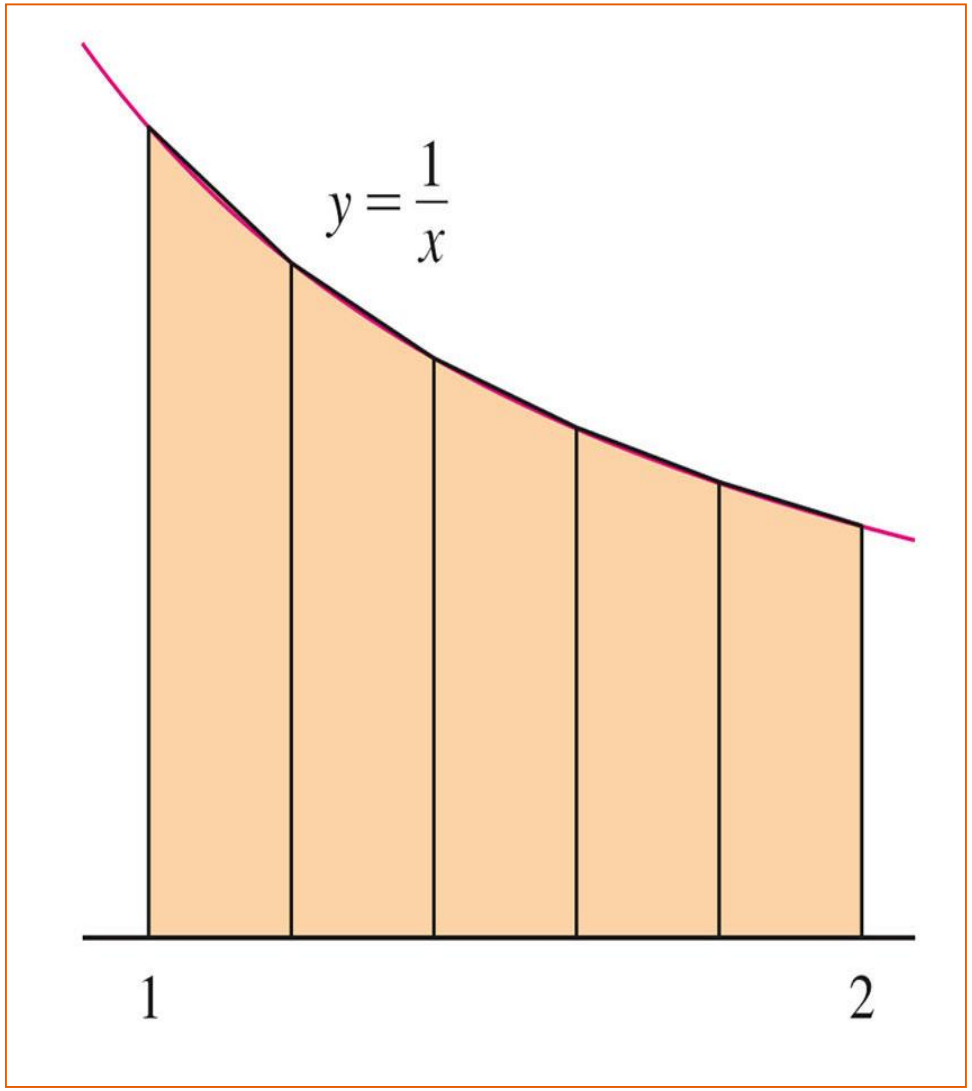
  - The error is |(0.693147 – 0.745639)*100/0.693147= 7.57%

APPROXIMATE INTEGRATION

Example 1(a)

The L_5 approximation is illustrated here.





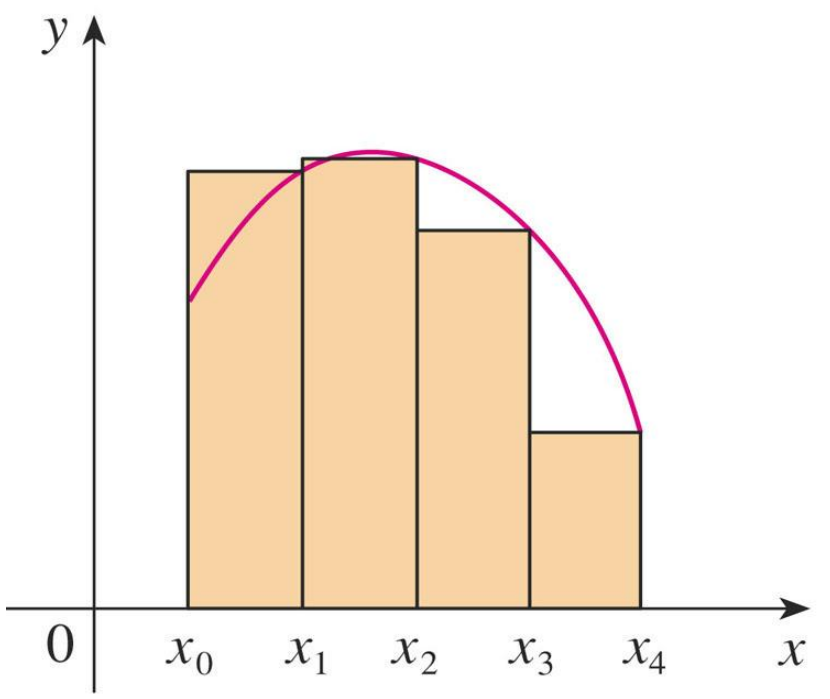
R_n APPROXIMATION

- If we choose x_i^* to be the right endpoint, $x_i^* = x_i$ and we have:

$$\int_a^b f(x)\,dx \approx R_n = \sum_{i=1}^{n} f(x_i)\,\Delta x$$

Equation-2

- The approximation R_n is called right endpoint approximation.



(b) Right endpoint approximation

APPROXIMATE INTEGRATION

Example 1(a)

- Approximate the integral of $f(x) = 1/x$ with $n = 5$, using right end-point approximation: $\int_1^2 (1/x)dx$

- With $n = 5$, $a = 1$ and $b = 2$, we have: $\Delta x = (2 - 1)/5 = 0.2$
  - <u>So, R_n the right endpoint approximation is :</u>

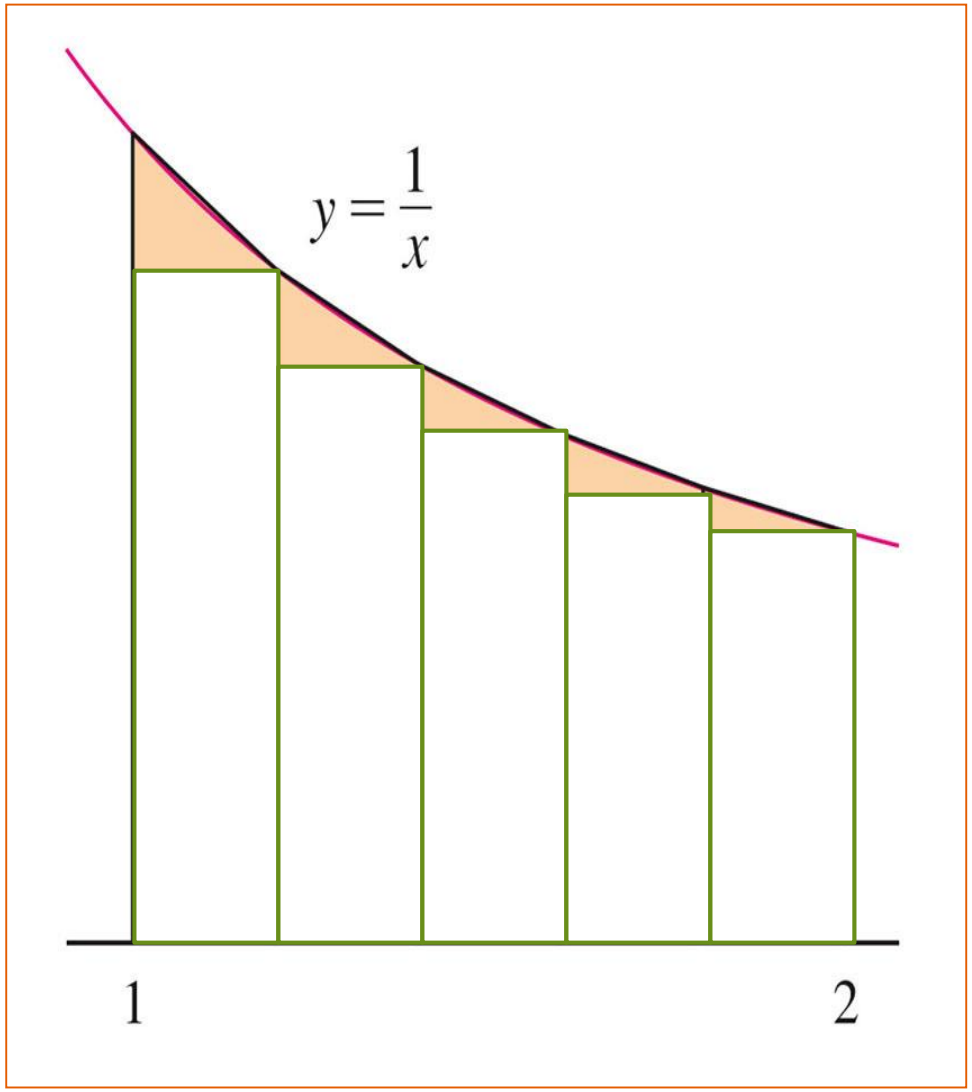
$$\int_1^2 \frac{1}{x}dx \approx R_5 = [f(1.2) + f(1.4) + f(1.6) + f(1.8) + f(2.0)] \times 0.2$$

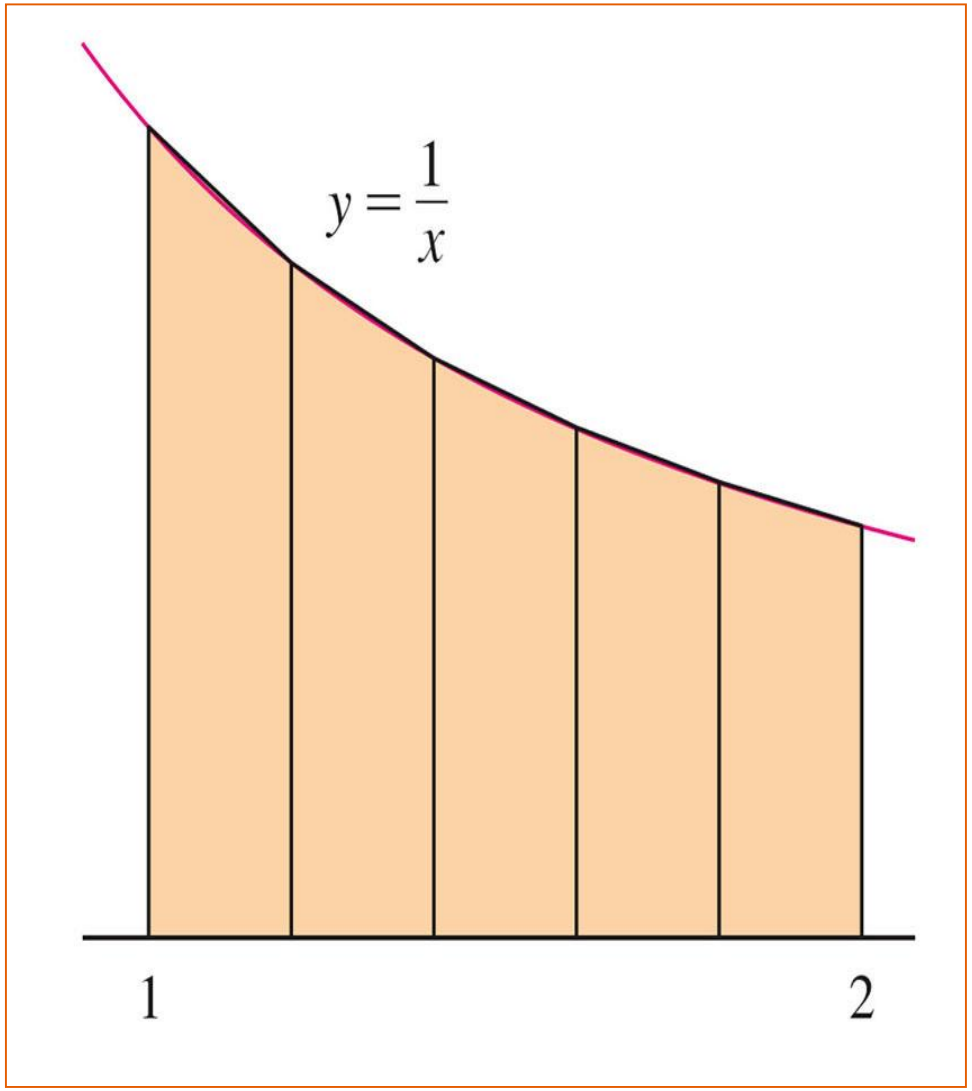
$$= 0.2\left(\frac{1}{1.2} + \frac{1}{1.4} + \frac{1}{1.6} + \frac{1}{1.8} + \frac{1}{2.0}\right) = 0.645634921(0.693147)$$

APPROXIMATE INTEGRATION

Example 1(a)

The R_5 approximation is illustrated here.

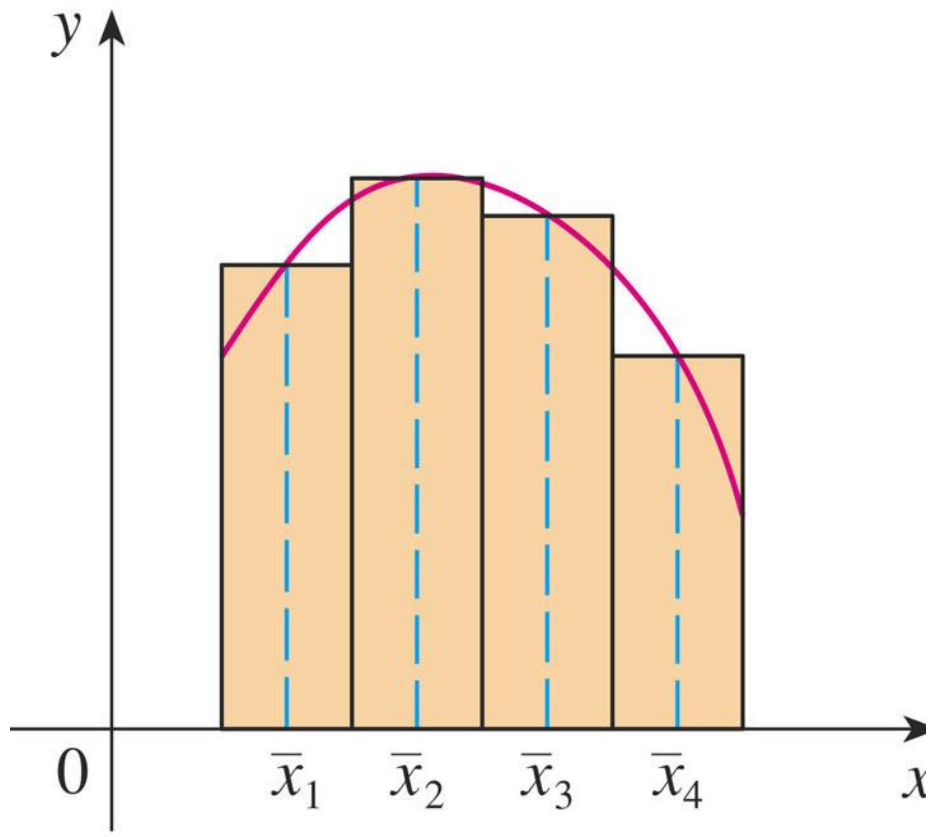




M_n APPROXIMATION

- We also considered the case where x_i^* is chosen to be the midpoint $\bar{x}_i = (x_{i-1} + x_i)/2$ of the subinterval $[x_{i-1}, x_i]$.

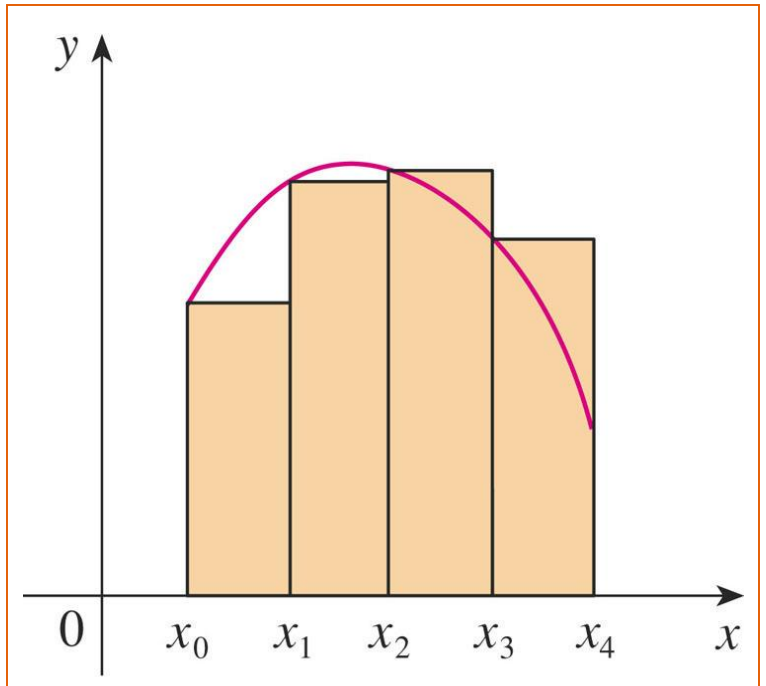
- The figure shows the midpoint approximation M_n.



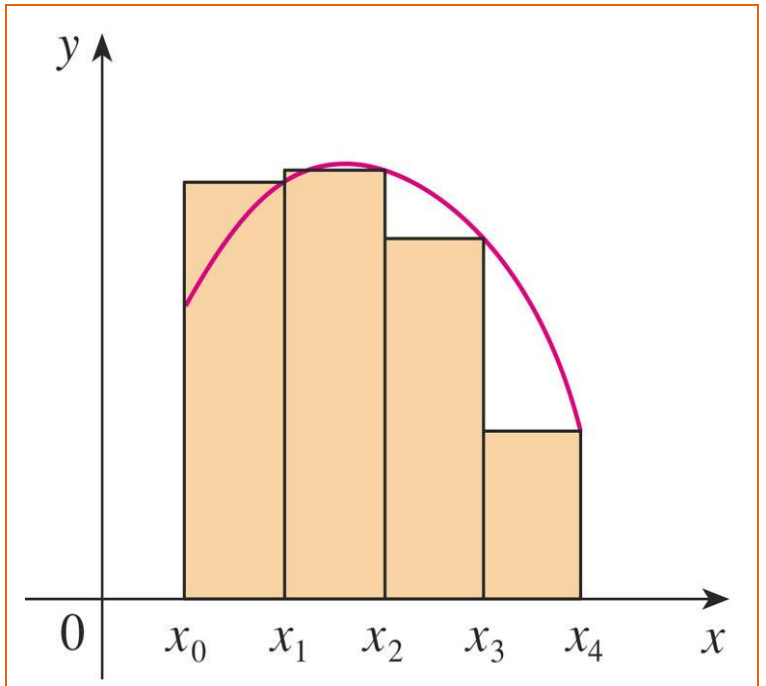
(c) Midpoint approximation

M_n APPROXIMATION

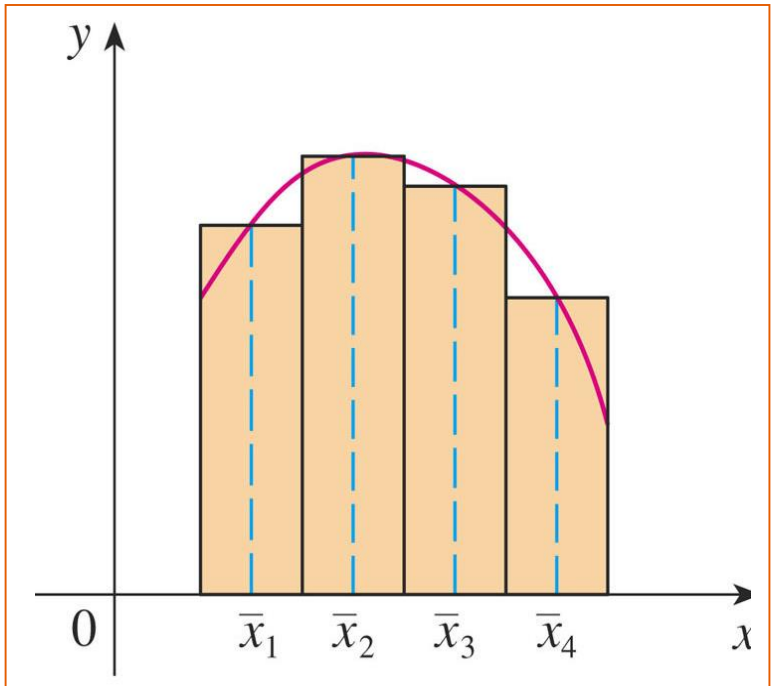
- M_n appears to be better than either L_n or R_n.



(a) Left endpoint approximation



(b) Right endpoint approximation



(c) Midpoint approximation

THE MIDPOINT RULE

$$\int_a^b f(x)\,dx \approx M_n$$

$$= \Delta x[\,f(\bar{x}_1) + f(\bar{x}_2) + \ldots + f(\bar{x}_n)]$$

- where $\Delta x = \dfrac{b-a}{n}$

- and $\bar{x}_i = \frac{1}{2}(x_{i-1} + x_i) = \text{midpoint of } [x_{i-1},\, x_i]$

APPROXIMATE INTEGRATION

Example 1(a)

- Approximate the integral of $f(x) = 1/x$ with $n = 5$, using mid-point approximation:

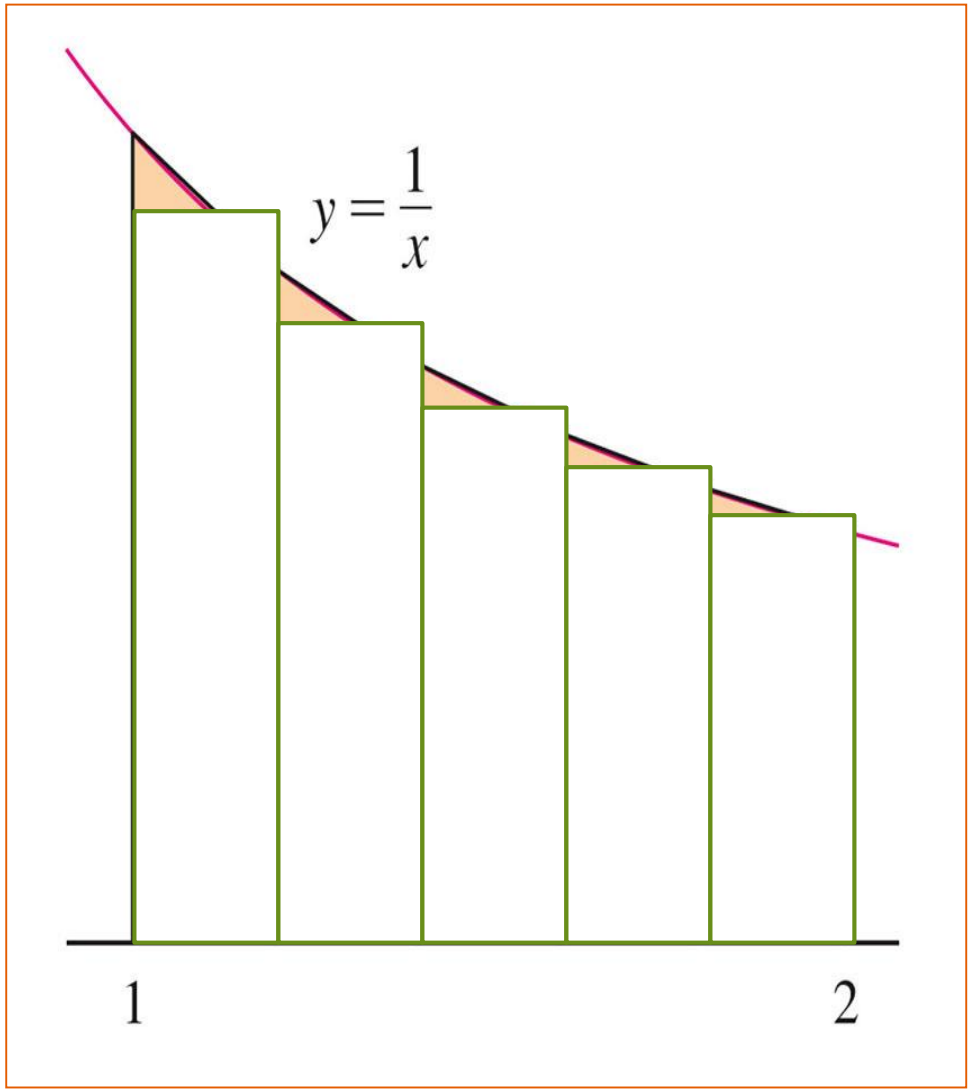
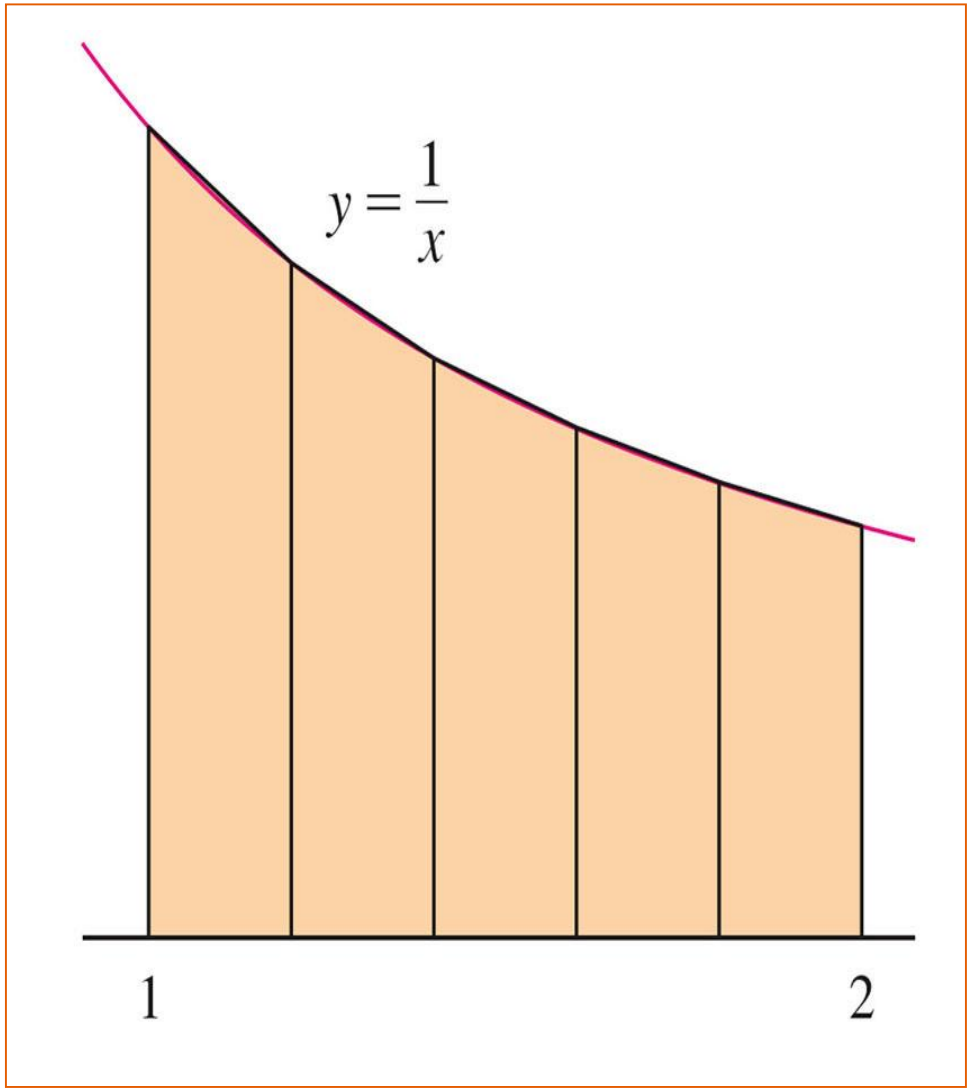
$$\int_1^2 (1/x)dx$$

- With $n = 5$, $a = 1$ and $b = 2$, we have: $\Delta x = (2 - 1)/5 = 0.2$
  - <u>So, R_n the mid-point approximation is :</u>

$$\int_1^2 \frac{1}{x}dx \approx R_5 = [f(1.1) + f(1.3) + f(1.5) + f(1.7) + f(1.9)] \times 0.2$$

$$= 0.2\left(\frac{1}{1.1} + \frac{1}{1.3} + \frac{1}{1.5} + \frac{1}{1.7} + \frac{1}{1.9}\right) = 0.691907886\ (0.693147)$$

APPROXIMATE INTEGRATION

Example 1(a)

The M_5 approximation is illustrated here.